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Experiment No. 4**Study of I-V Characteristics of Linear Circuits****Objective**

The aim of this experiment is to acquaint students with the concept of I-V characteristics. They will find I-V characteristics of some linear components and some circuits consisting linear combinations of them.

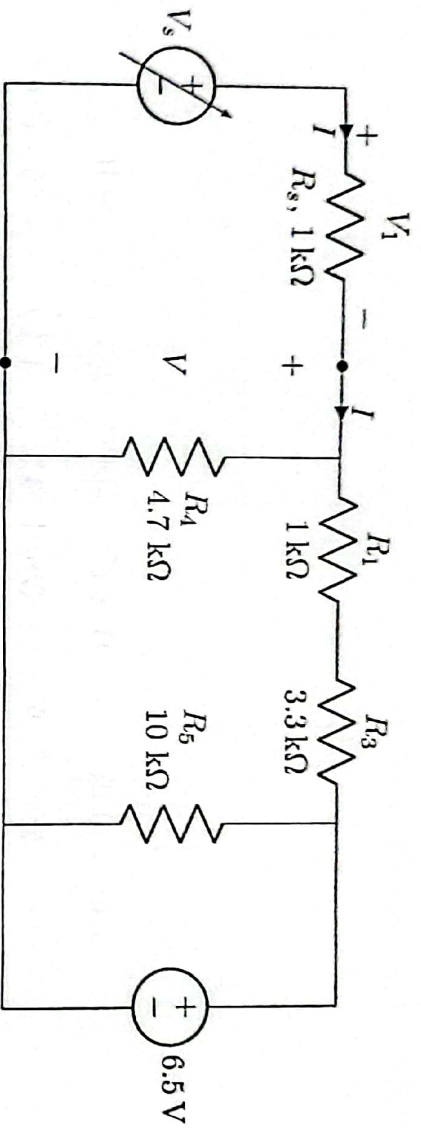
Part 1: By Using Multimeter**Theory**

I-V characteristics, also known as current-voltage characteristics, describe the relationship between the current flowing through a device/circuitry and the corresponding potential difference (voltage) across it. This concept is commonly used in the field of electronics and electrical engineering to analyze the behavior of various components such as resistors, diodes, transistors, and in general, circuits.

I-V characteristics provide a way to understand how current and voltage interact in electrical and electronic components and circuits. By analyzing these characteristics, circuits/devices can be designed and optimized, appropriate components can be selected, and the behavior of devices under different operating conditions can be predicted.

In hardware labs, studying the I-V characteristics of an element/circuitry can be done in some simple steps. After building the circuit using hardware tools (such as Breadboards, Power Supply), a multimeter or other measuring instruments can be used to measure the voltage and current at specified terminals in the circuit. The multimeter probes can be placed across the component or along the desired path (specified by the terminals) to measure the voltage difference and current flow. To determine the I-V characteristics, the voltage or current across the circuit or specific components must be varied. This can be done by adjusting the power supply voltage, using variable resistors, or changing the values of other circuit parameters. As the voltage or current is varied and the corresponding values are measured, the data can be recorded in a table. The voltage and current values for each point of interest in the circuit should be noted.

Once we have the values of currents and voltages at various points in the circuit, the I-V characteristics can be plotted. Typically, this involves creating a graph with current (I) on the y-axis and voltage (V) on the x-axis.



Circuit 3

- > For each of these circuits, apply the specified supply voltages (from the first column of their respective data tables) using the DC power supply.
- > Measure the voltage, V_1 across the $1\text{ k}\Omega$ resistor using the multimeter and use Ohm's law to calculate the current I through the two terminals (denoted by • in the circuits).
- > Measure the voltage, V across the two terminals (denoted by • in the circuits) using the multimeter, and fill up the data tables.

Data Tables

Signature of Lab Faculty:

S. Hussain

Date:

22.6.23

**** For all the data tables, take data up to three decimal places, round to two, then enter into the table.**

Table 1: Resistance Data

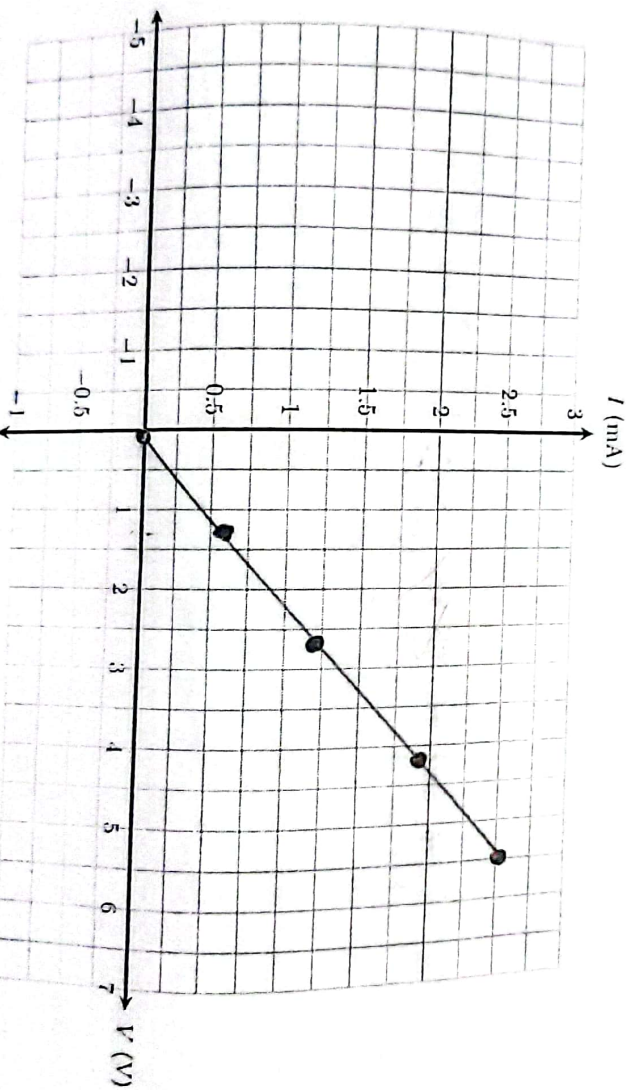
For all your future calculations, please use the observed values only (even for theoretical calculations).

Notation	Expected Resistance	Observed Resistance ($\text{k}\Omega$)	Notation	Expected Resistance	Observed Resistance ($\text{k}\Omega$)
R_s	$1\text{ k}\Omega$	0.999	R_3	$3.3\text{ k}\Omega$	3.261
R_1	$1\text{ k}\Omega$	0.985	R_4	$4.7\text{ k}\Omega$	4.60
R_2	$2.2\text{ k}\Omega$	2.163	R_5	$10\text{ k}\Omega$	9.94

Table 2: Data from Circuit 1

V_s (V)	V_1 (V)	V (V)	$I = \frac{V}{R_s}$ (mA)	
			Exper- imental	Theo- retical
Expected Voltage	From DC power supply	Using multi- meter	Exper- imental	Theo- retical
0.0	0.0	0.005	-0.0011	0
2.0	2.0	2.009	0.63	0.64
4.0	4.0	4.02	1.256	1.28
6.0	6.0	6.08	1.898	1.92
8.0	8.0	8.04	2.507	2.55
			Exper- imental	Theo- retical
			-0.0025	0
			0	-0.0011
			0	0
			1.37	0.636
			2.80	1.26
			4.22	1.917
			5.63	2.56

Plot the values of I and V from the above table.



Draw the best-fitting straight line through all the data points.

Slope of the straight line, $m =$

0.455 $\text{k}\Omega^{-1}$

Resistance from the plot, $R_T = \frac{1}{m} =$

2.197 $\text{k}\Omega$

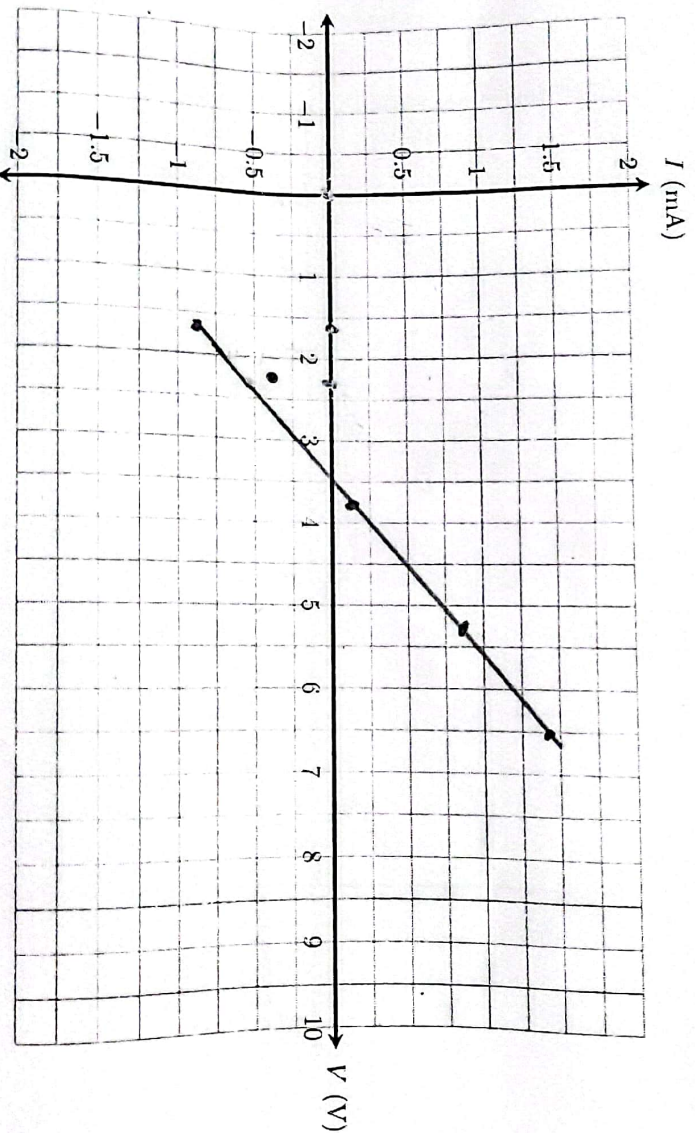
Percentage of Error = $\left| \frac{R_2 - R_T}{R_2} \right| \times 100\% =$

1.57 %

Table 3: Data from Circuit 2

Expected Voltage	V_s (V)	V_1 (V)		V (V)		$I = \frac{V}{R_t}$ (mA)	
		Using multi-meter	Experimental	Theoretical	Experimental	Theoretical	Theoretical
0.0	0.8	0.785 0.785	-0.813	-0.911	1.601	1.71	-0.821 -0.90
2.0	2.0	2.019	-0.431	-0.499	2.450	2.17	-0.445 -0.42
4.0	4.0	4.03	0.191	0.10	3.83	3.49	0.192 0.10
6.0	6.0	6.08	0.820	0.93	5.22	5.64	0.828 0.72
8.0	8.0	8.06	1.453	1.50	6.62	6.95	1.467 1.51

Plot the values of I and V from the above table.



Draw the best-fitting straight line through all the data points.

Slope of the straight line, $m =$

0.457 $\text{k}\Omega^{-1}$

Resistance from the plot, $R_T = \frac{1}{m} =$

2.185 $\text{k}\Omega$

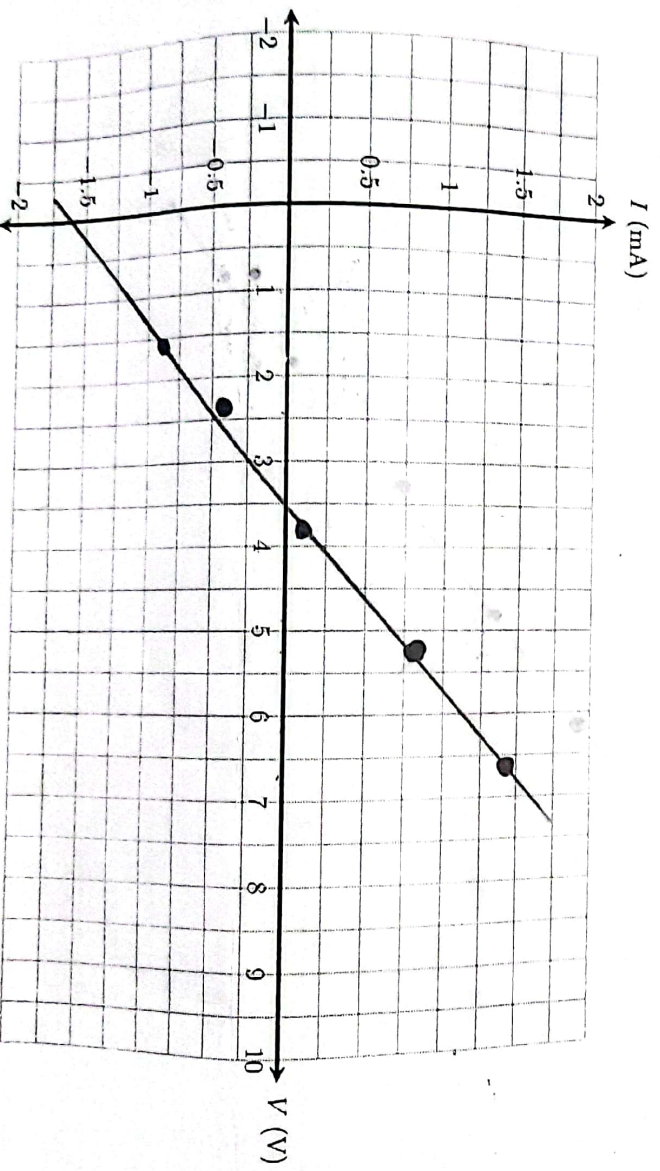
The straight line intersects x-axis at, $V_T =$

3.4 V

Table 4: Data from Circuit 3

Expected Voltage	V_s (V)	From DC power supply	Using multi-meter	V_1 (V)		V (V)		$I = \frac{V_1}{R_s}$ (mA)	
				Experi-mental	Theo-retical	Experi-mental	Theo-retical	Experi-mental	Theo-retical
0.0	0.7	0.7	0.77	0.81	-0.72	4.88	1.71	-0.82	-0.92
2.0	2.0	2.0	2.04	2.02	-0.47	2.46	2.20	-0.42	-0.50
4.0	4.0	4.0	4.08	0.18	0.9	3.84	3.42	0.18	0.11
6.0	6.0	6.0	6.07	0.82	0.93	5.27	5.65	0.83	0.92
8.0	8.0	8.0	8.04	1.17	1.52	6.61	6.87	1.44	1.51

Plot the values of I and V from the above table.



Draw the best-fitting straight line through all the data points.

Slope of the straight line, $m =$

0.4599 kΩ⁻¹

Resistance from the plot, $R_r = \frac{1}{m} =$

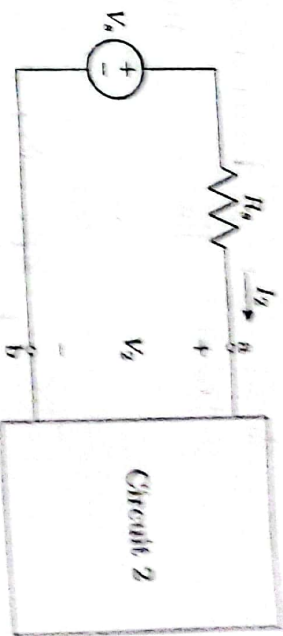
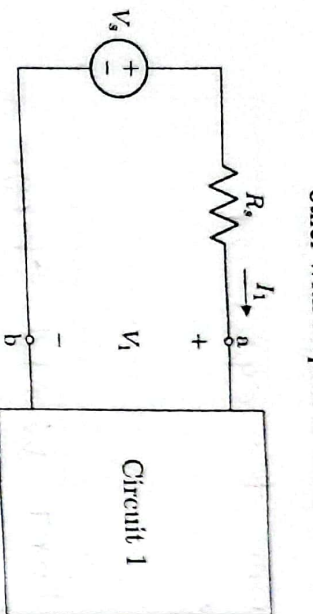
2.193 kΩ

The straight line intersects x-axis at, $V_r =$

3.5 V

Questions

1. What conditions must exist for the following two circuits to be equivalent to each other with respect to terminals a - b?



If $V_1 = V_2$ and $I_1 = I_2$. Also the circuits will be equivalent.

2.

- (a) For the Circuit 2 you constructed in the laboratory, derive a relation between I and V .

$$\begin{aligned}
 -V + I + 2 \cdot 2I + 3 \cdot 4 &= 0 \\
 \Rightarrow I &= \frac{V_3 - 3 \cdot 4}{1 + 2 \cdot 2} \\
 &= \frac{V_3 - 3 \cdot 4}{3 \cdot 2}
 \end{aligned}$$

- (b) For the Circuit 3 you constructed in the laboratory, derive a relation between I and V .

In this circuit, there is an additional circuit attached with 6V voltage source. That why relation between circuit 2 and circuit one same.

$I = mV + c$

$m = \text{slope}$
 $c = \text{constant}$

$I = 0.45 \times 1.56$
 if $V = 6.07$, $I = (0.45 \times 6.07) - 1.52 = 1.26 \text{ mA}$

(c) Did you notice any similarity between the $I - V$ relationships in (a) and (b)?

☒ Yes ☐ No

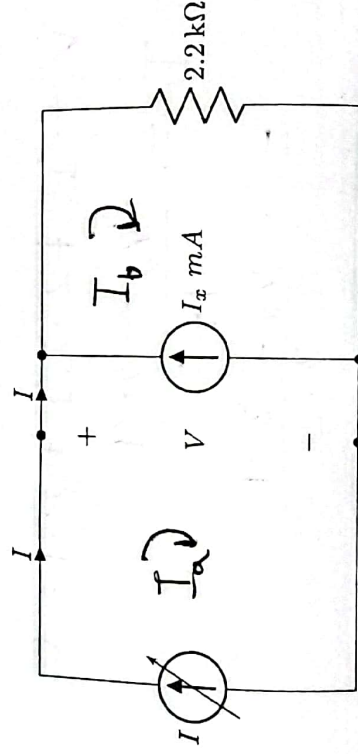
If yes, what are they?

Both are founded from the equ. of straight line, $y = mx + c$

(d) Will it have any effect if one of these two circuits is replaced with the other? Why?

The slope and constant would change gradually, when it is replaced.

(e) Now, for the following circuit, determine the value of I_x so that the $I - V$ relation matches with those you derived in (a) and (b). Is this circuit also equivalent to Circuit 2 and Circuit 3?



KVL loop - 1:

$$I_a + V = 0 \quad \text{--- (i)}$$

Mesh analysis loop - 2:

$$\text{--- } V + 2.2 I_b = 0 \quad \text{--- (ii)}$$

iii

$$I_1 + 2.2 I_2 = 0$$

$$\therefore \text{super mesh} \Rightarrow I_b - I_a = I_x \Rightarrow I_x = 3.2 I_2$$

3. Can you think of any way where voltage-axis intersecting points (V_T) could be measured directly for Circuit 1 and Circuit 2?

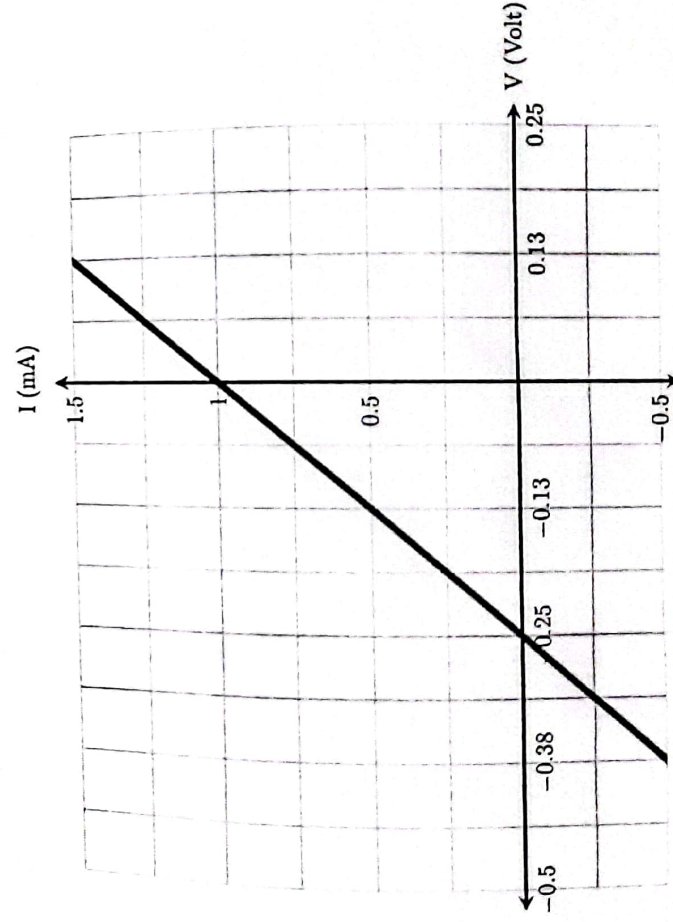
circuit 1: To determine voltage drop, a Voltmeter can be used.

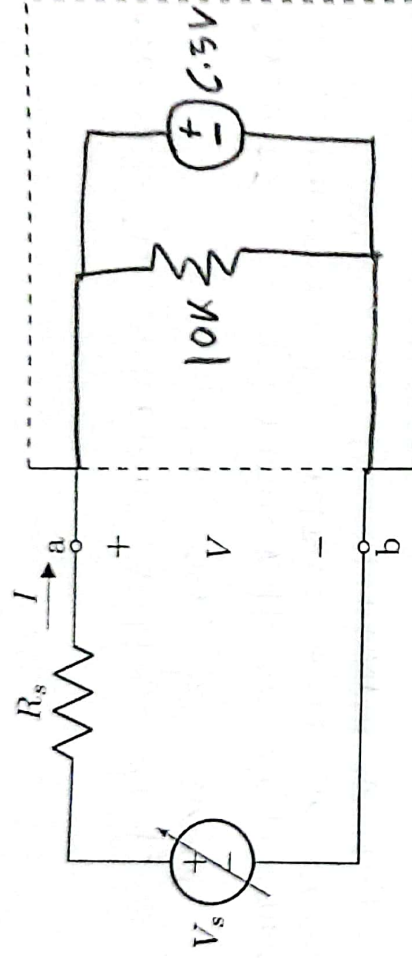
circuit 2: To determine voltage drop connect a voltmeter. The voltage axis intersecting point occurs at the forward voltage drop.

4. In general, what is the simplest technique to derive the $I - V$ characteristics of any linear two terminal circuit? How many minimum data points are required?

Plotting $I - V$ on a graph is the simplest way to find out $I - V$ characteristics. Minimum two data points ~~are~~ required.

5. A linear two-terminal circuit has the following $I - V$ relationship at the terminals $a - b$ measured in a laboratory with the setup shown below. Draw (in the next page) a simplified version of the circuit that can give rise to the same $I - V$ as shown. Determine the corresponding parametric values of the circuit elements.





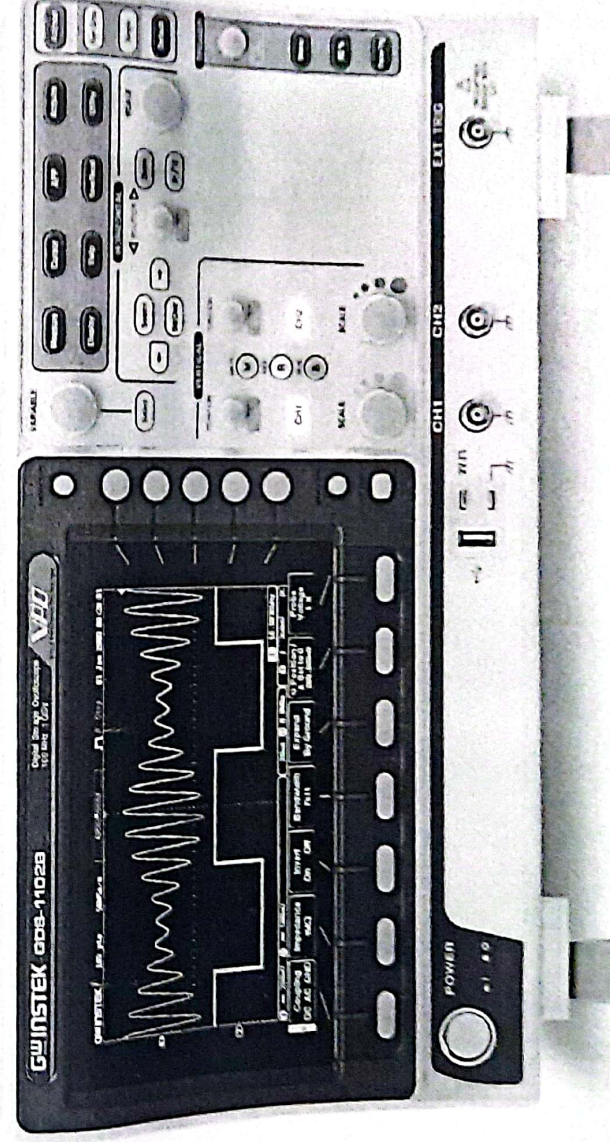
Draw inside the dashed box

Part 2: By Using Oscilloscope

Theory

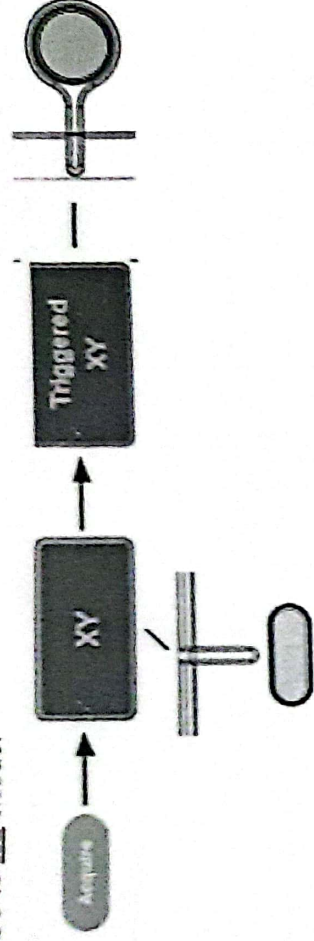
Oscilloscope

Oscilloscope is a device that can measure a sequence of voltages over time and can display that information by plotting them on a screen. In fact, oscilloscopes available at our labs are dual channel (CH1 and CH2), meaning, they can simultaneously show voltage vs time graph across two separate set of nodes.

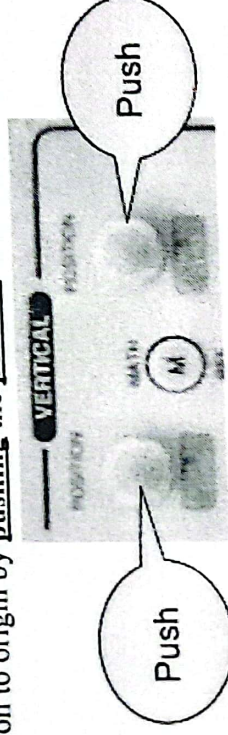


An oscilloscope

➤ Go to **XY** Mode:



➤ Set position to origin by pushing the position knobs on each channel.



➤ Turn both channel knobs so that the voltage resolutions are at 2V per division.



Questions

6.

(a) In normal mode of operation, an oscilloscope always plots -

☒ voltage as a function of time

☐ current as a function of time

(b) In X-Y mode of operation, an oscilloscope plots Channel-1 along the -

☒ y-axis ☐ x-axis

(c) Is there any way to observe the one you haven't selected in (a)?

1) current probe: It is used to measure high current signal

ii) shunt Resistor: To measure Voltage drop shunt can be connected in series with the circuit.

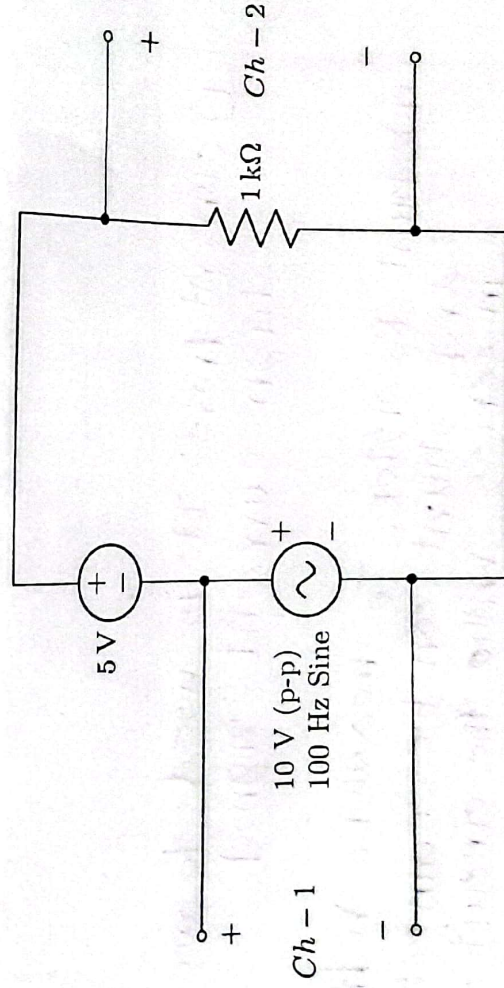
7. Put a checkmark beside the correct answers:

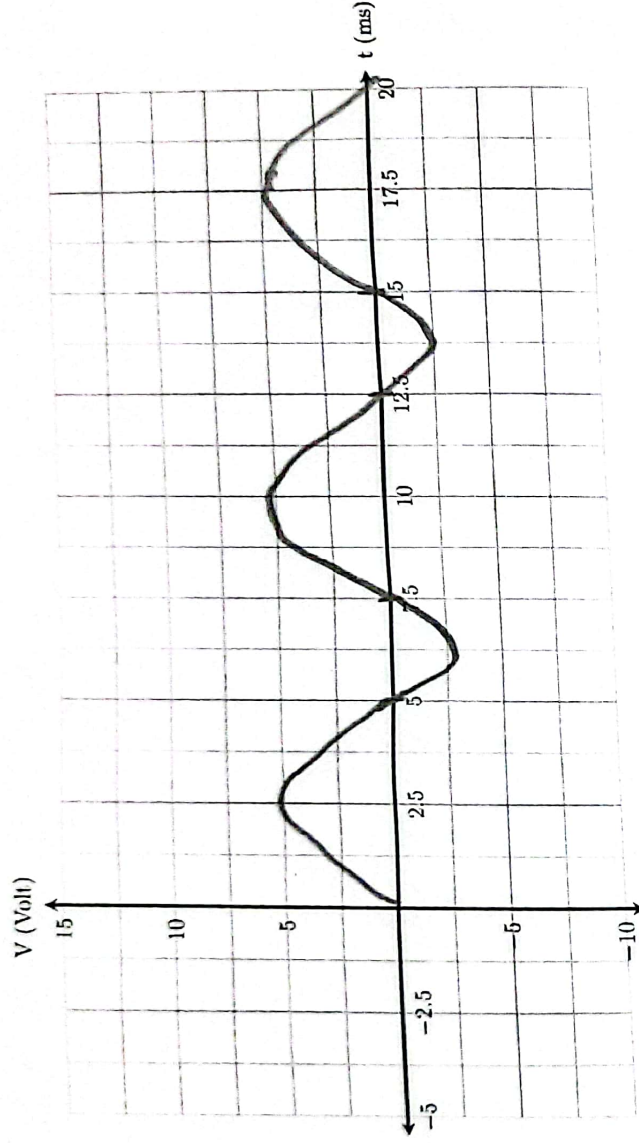
- a. The I-V characteristics of the following circuits were **straight lines** -
☒ Circuit 4 ☒ Circuit 5 ☒ Circuit 6
- b. The I-V characteristics of the following circuits went **through origin** -
☐ Circuit 4 ☒ Circuit 5 ☒ Circuit 6
- c. The following circuits were **equivalent to a resistor** -
☐ Circuit 4 ☒ Circuit 5 ☒ Circuit 6
- d. When the LDR was completely **in darkness**, the I-V characteristic line was -
☐ y-axis ☒ x-axis ☐ parallel to x-axis but shifted upwards
- e. When the LDR was completely **in darkness**, it was equivalent to -
☐ short circuit ☒ open-circuit ☐ 1 k Ω resistor ☒ 0A current source

8. Why was it necessary to invert Channel-2 of the Oscilloscope in order to visualize the *I vs. V* plot of Circuit 4, 5, and 6?

We used inverter to change the orientation of the signal voltage on the 4 signal input connector. To stabilize wave form & display char 2. in inverted - as voltage source of circuit 4 in properly aligned with triggering of oscilloscope

9. Draw the waveforms that should be observed in Channel-1 and Channel-2 of an oscilloscope when both the channels are ON and are connected in a setup shown below. Draw both the plots in the same template given below. Mark the waveforms according to their visualizing channel.





Report

1. Fill up the theoretical parts of all the data tables.
2. Answer to the questions.
3. Attach the captured images of the $I - V$ plot observed in oscilloscope for Circuits 4, 5, and 6.
4. Discussion [comment on the obtained results and discrepancies]. Start writing from below the line.

We have found the obtained results satisfactory. There was some discrepancies while using oscilloscope but the end result was fine. The $I - V$ plot line was bit shaky. Other than that we could finish our tasks very efficiently, though in table 4 we put some wrong data and again had to check and find the right value.

GW INSTEK GDS-1102B

Digital Storage Oscilloscope
100 MHz 1 GS/s

VPO
Visual Persistence Oscilloscope

HARD COPY

VARIABLE

Measure

Cursor

APP

Acquire

Display

Help

Save/Recall

Utility

HORIZONTAL

POSITION

SCALE

Select

Search

Set/Clear

Zoom

/H

VERTICAL

POSITION

POSITION

MATH

M

REF

R

BUS

B

SCALE

SCALE

CH1

CH2

EXT TRIG

POWER

2V Ω

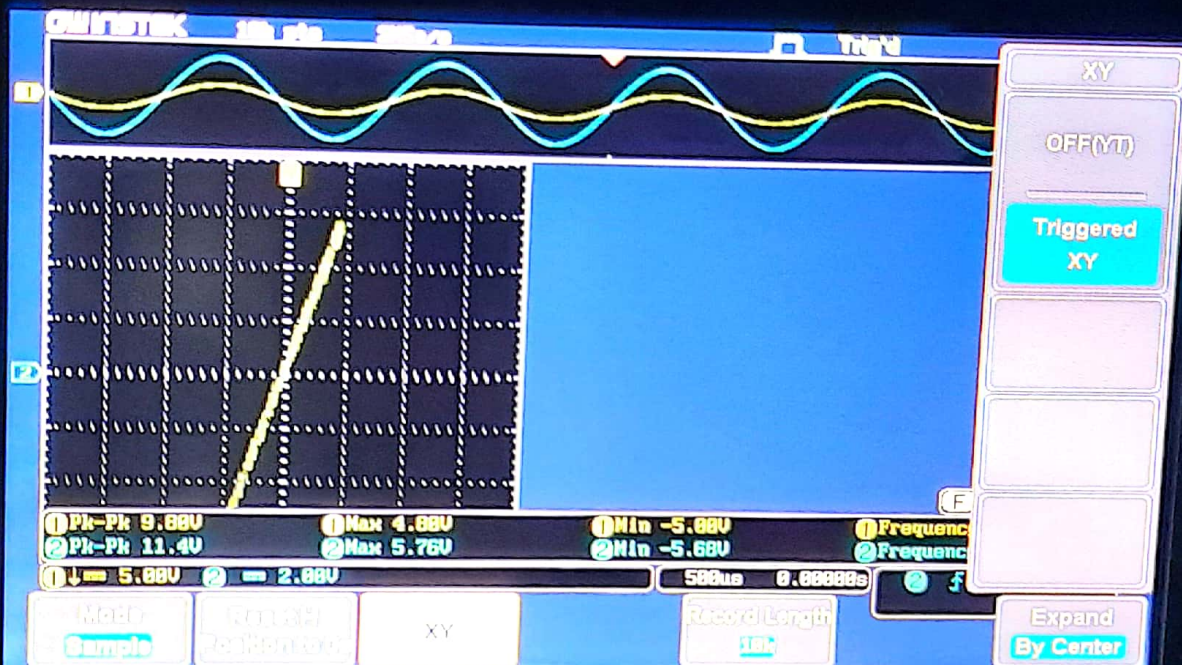
MAX

MAX

GW INSTEK GDS-1102B

Digital Storage Oscilloscope
100 MHz 1 GS/s

VPO
Visual Persistence Oscilloscope



VARIABLE

Measure

Cursor

Display

Help

HORIZONTAL

Select

Search

Set/Clare

VERTICAL

POSITION

POSITION

MATH

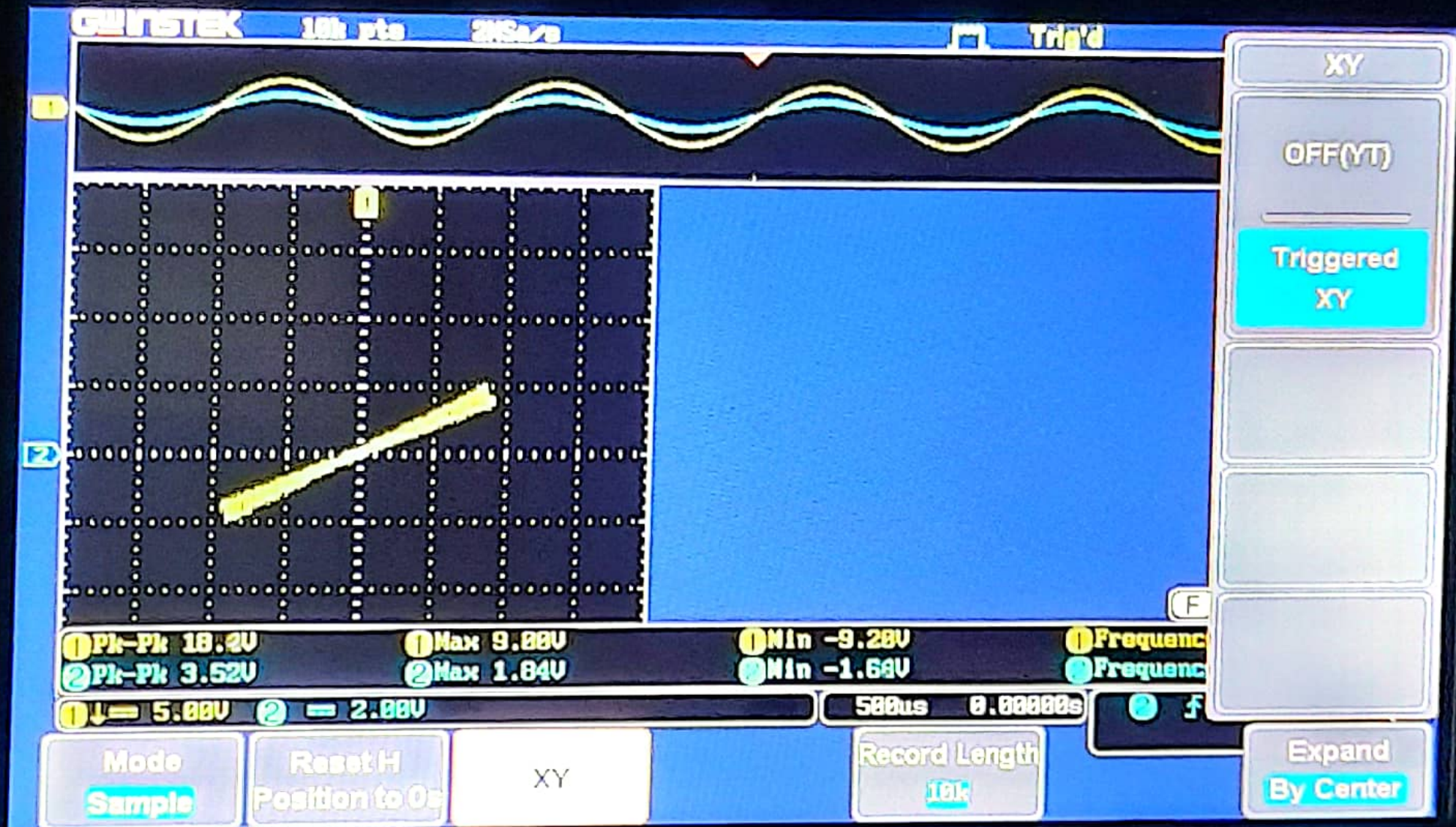
CH1

CH2

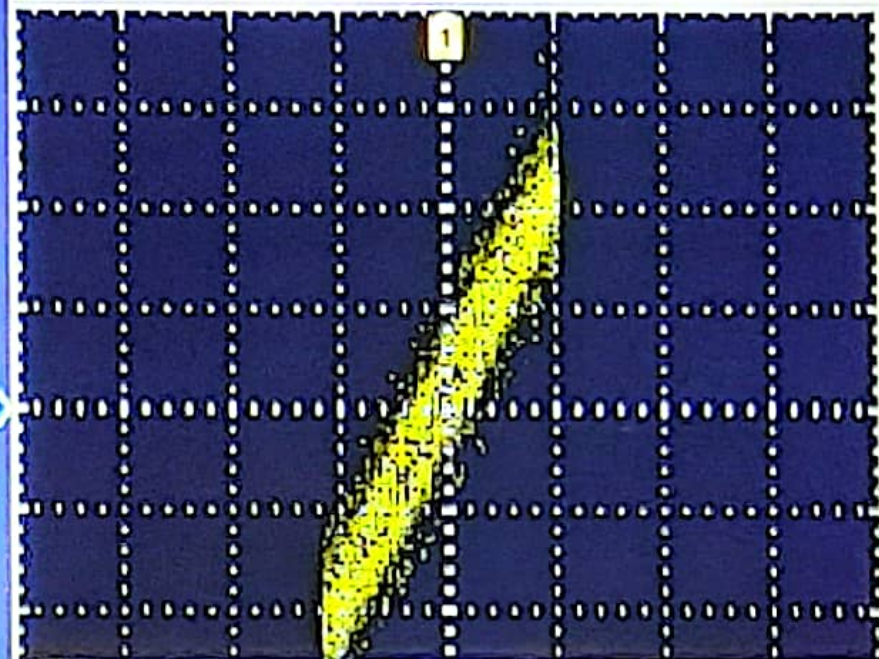
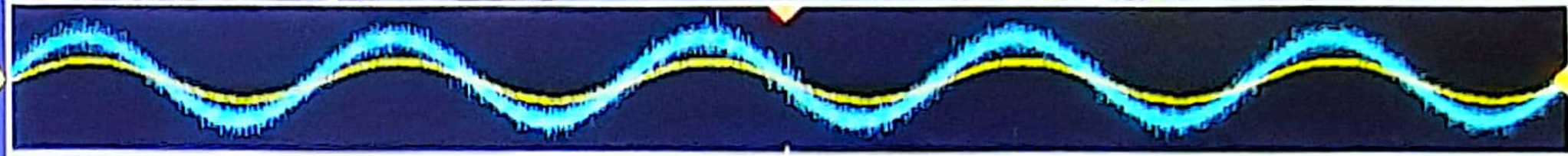
GW INSTEK GDS-1102B

Digital Storage Oscilloscope
100 MHz 1 GS/s

VPO
Visual Persistence Oscilloscope



10k pts 2MSa/s



① Pk-Pk 21.2V	① Max 18.0V	① Min -11.2V	① Frequency 3.000kHz
② Pk-Pk 588nV	② Max 388nV	② Min -288nV	② Frequency 1.012kHz
① ↓ = 18.0V	② = 188nV	588us 0.00000s	① f 8V DC

Coupling DC AC GND	Impedance 1MΩ	Invert On Off	Bandwidth Full	Expand By Ground	Position / Set to 0 0.000V	Probe Voltage 1 X
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